PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
rules = [
    ({"fever", "cough"},                "viral_infection"),
    ({"viral_infection", "body_pain"},  "flu"),
    ({"fever", "chills", "sweating"},   "malaria"),
    ({"fever", "stomach_pain", "weakness"}, "typhoid"),
    ({"high_fever", "rash", "joint_pain"},  "dengue"),
    ({"sneezing", "runny_nose"},        "common_cold"),
]
symptoms = {"fever", "cough", "body_pain"}
diseases = ["flu", "malaria", "typhoid", "dengue", "common_cold"]
def prove(goal, depth=0):
    pad = "    " * depth
    if goal in symptoms:
        print(f"{pad}{goal}: patient has it  [TRUE]")
        return True
    for conditions, conclusion in rules:
        if conclusion == goal:
            print(f"{pad}Goal '{goal}' needs {sorted(conditions)}")
            if all(prove(c, depth + 1) for c in sorted(conditions)):
                print(f"{pad}'{goal}' is proved  [TRUE]")
                return True
            print(f"{pad}'{goal}' failed  [FALSE]")
            return False
    print(f"{pad}{goal}: not a symptom of patient  [FALSE]")
    return False
print("Patient symptoms:", sorted(symptoms))
found = []
for d in diseases:
    print(f"\n--- Checking goal: {d} ---")
    if prove(d):
        found.append(d)
print("\nDiagnosis:", ", ".join(found) if found else "No disease proved. Consult a doctor.")
```

OUTPUT:

```
>>>
    ==================== RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ====================
    Patient symptoms: ['body_pain', 'cough', 'fever']

    --- Checking goal: flu ---
    Goal 'flu' needs ['body_pain', 'viral_infection']
        body_pain: patient has it  [TRUE]
        Goal 'viral_infection' needs ['cough', 'fever']
            cough: patient has it  [TRUE]
            fever: patient has it  [TRUE]
        'viral_infection' is proved  [TRUE]
    'flu' is proved  [TRUE]

    --- Checking goal: malaria ---
    Goal 'malaria' needs ['chills', 'fever', 'sweating']
        chills: not a symptom of patient  [FALSE]
    'malaria' failed  [FALSE]

    --- Checking goal: typhoid ---
    Goal 'typhoid' needs ['fever', 'stomach_pain', 'weakness']
        fever: patient has it  [TRUE]
        stomach_pain: not a symptom of patient  [FALSE]
    'typhoid' failed  [FALSE]

    --- Checking goal: dengue ---
    Goal 'dengue' needs ['high_fever', 'joint_pain', 'rash']
        high_fever: not a symptom of patient  [FALSE]
    'dengue' failed  [FALSE]

    --- Checking goal: common_cold ---
    Goal 'common_cold' needs ['runny_nose', 'sneezing']
        runny_nose: not a symptom of patient  [FALSE]
    'common_cold' failed  [FALSE]

    Diagnosis: flu
>>>
```